Abhisek Singh

AI Architect / Senior AI Engineer: Agentic AI · LLM Systems in Production · RAG · MCP · Distributed Systems

Bengaluru, India · raizadaabhi11@gmail.com · linkedin.com/in/abhisek-singh-952738b8 · github.com/abhisek11 · abhisek11.github.io

SUMMARY

AI engineer and architect who designs and ships agentic AI and LLM systems to production, building on 8+ years of Python backend and distributed-systems engineering. As a Technical Lead at GlobalLogic on Verizon's VzCloud platform, I took multiple agentic systems from proof of concept to production with LangGraph orchestration, MCP tool interfaces, RAG over pgvector, knowledge-graph planning, and evaluation and guardrail release gates. I set AI/RAG architecture standards used across teams. I also designed, built and run RoadLink, a live multi-tenant SaaS. Previously, I was an Engineering Manager at PayMob (12 engineers, $200M monthly payment volume).

SKILLS

- **Agentic AI & LLMs:** LangGraph, CrewAI, AutoGen, MCP servers and tool design, RAG (pgvector), knowledge graphs (Apache AGE), Pydantic structured outputs, prompt engineering, evaluation harnesses, guardrails, human-in-the-loop, multi-model consensus; OpenAI, Anthropic Claude, Google Gemini
- **Languages:** Python, Java, Go, SQL, JavaScript, Dart
- **Backend:** FastAPI, Django / DRF, Celery, REST and GraphQL, microservices, Kafka, Temporal.io, multi-tenancy, OAuth/JWT, RBAC
- **Data & cloud:** PostgreSQL, pgvector, ClickHouse, Redis, AWS (EC2, RDS, S3, Lambda), Docker, Kubernetes, CI/CD, OpenTelemetry, Prometheus, Grafana
- **Leadership:** managed 12 engineers, HLD/LLD reviews, mentoring, stakeholder and roadmap negotiation

EXPERIENCE

Technical Lead, GlobalLogic (client: Verizon) · **Apr 2024 – Present · Bengaluru**

- Took multiple agentic AI systems from PoC to production on VzCloud: LangGraph orchestration, MCP tool interfaces, RAG over pgvector, and knowledge-graph planning on PostgreSQL/Apache AGE, with evaluation and guardrails gating each stage.
- Designed Pydantic-validated tool interfaces and structured-output contracts, with human-in-the-loop checks and auth/RBAC on every tool call.
- Set AI/RAG, API, database and distributed-systems standards used across teams. Lead architecture for Python microservices on AWS, Docker and Kubernetes, and own HLD/LLD reviews.
- Moved long-running workflows onto Temporal.io with retries, circuit breakers and tracing. Built Kafka event pipelines and real-time ClickHouse analytics over multi-billion-row datasets.
- Drive incident response (RCA, postmortems), carry on-call, and mentor engineers.

Founder & Architect, RoadLink (roadlink.co.in) · **2025 – Present**

- Designed, built solo and run a live multi-tenant SaaS for Indian road-freight fleets: dispatch, GST/TDS/TCS billing, per-truck ledgers, trip settlement, POD capture and GPS tracking.
- ~100k lines of Python (Django 5, PostgreSQL, Redis, Celery; 15 apps, 95 models) and ~41k lines of Dart (Flutter app for 4 user roles).

- Built two-model (OpenAI + Gemini) AI ledger reconciliation that runs deterministic checks first, lets models only propose changes, and makes every change one-click reversible. This cut a run from ~40 model calls to 1–2.
- Built shared-schema multi-tenancy with a strict cross-tenant boundary and 35+ RBAC permissions. Cut the production Docker image from 6.4 GB to 845 MB.

Engineering Manager, PayMob · **Nov 2021 – Dec 2023 · Remote**

- Led a 12-engineer backend team scaling a payments platform to $200M in monthly transaction volume.
- Built transaction, settlement and reconciliation services, KYC onboarding and fraud controls under PCI-grade security and full audit trails.
- Set engineering standards for API design, code review, testing and CI/CD, and grew several engineers into leads.

Software Engineer, McKinsey & Company engagements · **2021 · Gurugram**

- Built Python and Java backend services and REST APIs for predictive-modelling client engagements.

Earlier: Capital Numbers, Shyam Future Tech, NTT DATA · **Nov 2017 – Dec 2020**

- Python/Django and C++ APIs on AWS, Docker deployments, payment-gateway integrations and data pipelines.

SELECTED PROJECTS

- **Autonomous Trading Floor:** multi-agent system across 6 MCP servers and 44 tools, with evaluation gates on every action.
- **AgentPay:** payments infrastructure for AI agents built on Lithic virtual cards, a Go authorization engine, a policy engine and an immutable Postgres ledger.
- **Support & sales agents:** RAG-grounded agents with human escalation on confidence or policy failure.

CERTIFICATIONS & EDUCATION

- GL Certified AI Professional: Generative AI, GlobalLogic (Jul 2026) · Introduction to Generative AI, Google (Dec 2023)
- B.E., Electrical and Electronics Engineering, Visvesvaraya Technological University, Bengaluru (2013 – 2017)